

Performance Analysis of Deep Learning Models for Sentiment Classification

Abstract

This study explores the effects of optimizers, training epochs, and deep learning architectures on sentiment classification. The results suggest that Adagrad optimizer, moderate training duration, and convolutional architectures offer the best balance of accuracy and generalization for this task.

1. Introduction

Sentiment classification, a crucial task in natural language processing, involves categorizing text by sentiment polarity, with widespread applications in fields like customer feedback analysis and social media monitoring^{[1][2]}. The IMDB dataset, a benchmark for sentiment analysis, offers real-world movie reviews that challenge models to handle diverse linguistic patterns and generalize effectively. Recent advancements in deep learning have enabled automatic feature extraction, reducing reliance on manual efforts. This study evaluates the performance of six deep learning architectures, including FFNNs, CNNs, LSTMs, and BiLSTMs, on the IMDB dataset, identifying the most effective approach for sentiment analysis.

2. Methodology

2.1 Environment Selection

The experiments, conducted on Google Colab with an A100 High-RAM GPU, leveraged its superior speed, memory capacity, and specialized for deep learning, outperforming T4 and L4 GPUs designed for lighter workloads.

2.2 Dataset

The IMDB dataset consists of 50,000 movie reviews labeled as positive or negative. Preprocessing includes tokenization and padding, with randomly initialized embeddings representing words.

2.3 Experiments

This study systematically evaluated various deep learning models for sentiment classification on the IMDB dataset in three stages:

2.4 Optimizer Experiments

Models were trained using SGD, Adam, and Adagrad optimizers. Training and validation loss and accuracy were recorded to assess optimizer performance and convergence.

2.5 Epoch Variation

Using the Adam optimizer, models were trained for 5, 10, 20, and 50 epochs to examine the impact of training duration on loss and accuracy, identifying trends of overfitting or underfitting.

2.6 Model Comparison

Six architectures were compared, trained with Adam for 50 epochs using random embeddings:

1. FFNNs: Single, two, and three-layer feedforward networks with hidden dimensions of 500, 300, and 200.
2. CNN: Capturing local n-gram patterns with kernel sizes 1, 2, and 3.
3. LSTM and BiLSTM: Sequential models capturing long-term and bidirectional context.

2.7 Metrics comparison

Metrics including training, validation, and test loss and accuracy were used to evaluate and compare model effectiveness.

3. Results

All the code, data results, figures/graphs are available in the GitHub repository:

https://github.com/dymasius12/Performance_Analysis_of_Deep_Learning_Models_for_Sentiment_Classification/tree/Main

3.1 Optimizer Variation

Table 1: Performance Metrics of Two Optimizer Variations in Sentiment Classification

Optimizer	Test Loss	Test Accuracy (%)	Final Train Loss	Final Validation Loss	Final Train Accuracy (%)	Final Validation Accuracy (%)
SGD	0.6829541557	56.01102941	0.6811004931	0.68541034	56.2931765	54.77577685
Adam	0.5930154604	71.08775575	0.3095372681	0.6292568455	88.28206467	74.02895482
Adagrad	0.5719512112	73.27925192	0.3638336409	0.5647456065	85.17091372	73.43308618
Max value	0.6829541557	73.27925192	0.6811004931	0.68541034	88.28206467	74.02895482
Avg value	0.6159736091	66.79267903	0.4514904674	0.6264709307	76.58205163	67.41260595

From the table, Adagrad achieves the highest test accuracy and the lowest test loss, along with a low final validation loss, indicating better generalization and minimal overfitting. The evaluation of SGD, Adam, and Adagrad optimizers revealed distinct performance trends across training, validation, and test datasets. Adagrad demonstrated superior results, achieving the lowest test loss (0.572) and highest test accuracy (73.28%), outperforming Adam (0.593 test loss, 71.09% accuracy) and SGD (0.683 test loss, 56.01% accuracy). Adagrad also exhibited consistent convergence and robust validation accuracy, maintaining over 73% across epochs, compared to Adam's peak at 74.03% and SGD's maximum of 56.01%. These findings highlight Adagrad's adaptive learning strategy, which aligns well with the dataset and enables better generalization

compared to the momentum-based Adam optimizer and the less adaptive SGD. The analysis underscores Adagrad's effectiveness in balancing convergence and performance, making it the most reliable optimizer in this experiment.

3.2 Epoch Variation

Table 2: Performance Metrics Across Four Epoch Variations in Sentiment Classification

Epochs	Test Loss	Test Accuracy (%)	Final Train Loss	Final Validation Loss	Final Train Accuracy (%)	Final Validation Accuracy (%)
5	0.6046235281	68.45108696	0.5445974972	0.6561562939	71.99719761	61.49805794
10	0.5825599952	68.93142584	0.4572660417	0.5764580692	80.214579	69.61069915
20	0.5928624342	70.7081202	0.3134964604	0.6223519293	88.14275939	74.61599576
50	0.6117901377	70.405211	0.1579048042	0.9019668999	94.18583813	71.22175143
Max value	0.6117901377	70.7081202	0.5445974972	0.9019668999	94.18583813	74.61599576
Avg value	0.5979590238	69.623961	0.3683162009	0.6892332981	83.63509353	69.23662607

With 20 epochs, good test accuracy and low test loss were achieved, while 50 epochs showed overfitting despite high training accuracy. Experiments with 5, 10, 20, and 50 epochs using the Adam optimizer revealed the impact of training duration on performance. The model achieved its highest test accuracy (70.71%) and lower test loss (0.593) at 20 epochs compared to 50 epochs, balancing generalization and overfitting. While shorter training (5 and 10 epochs) yielded lower accuracies (68.45% and 68.93%, respectively), extended training to 50 epochs led to slight overfitting, evident from the increased validation loss (0.902) despite strong training accuracy (94.19%). These results underscore the importance of selecting an optimal number of epochs to maximize performance while minimizing overfitting.

3.3 Model Variation

Table 3: Performance Metrics for Six Model Variations in Sentiment Classification

Model	Test Loss	Test Accuracy (%)	Final Train Loss	Final Validation Loss	Final Train Accuracy (%)	Final Validation Accuracy (%)
OneLayerFFNN	0.6930241781	50.46059982	0.6931375907	0.6930801262	50.26259981	50.6950508
TwoLayerFFNN	0.6929576216	50.26699936	0.6930135958	0.6932912929	51.11514215	50.07585479
ThreeLayerFFNN	0.6929548201	50.54688426	0.6932630006	0.6929963044	50.28630571	50.41490407
CNN	0.6935093965	50.8561655	0.6975066899	0.7015848635	50.7838936	50.80429846
LSTM	0.6935136804	49.58207229	0.6933693636	0.6928956317	50.781065	50.94105202
BiLSTM	0.693351064	50.54507785	0.6933246403	0.6937328436	50.72052279	50.87766008
Max value	0.6935136804	50.8561655	0.6975066899	0.7015848635	51.11514215	50.94105202
Avg value	0.6932184601	50.37629985	0.6939358135	0.6945968437	50.65825484	50.63480337

Among the six tested architectures, the CNN model demonstrated the highest test accuracy (50.86%) but did not achieve the lowest test loss (0.6935), indicating moderate effectiveness in capturing local dependencies. Traditional feedforward networks (One-layer, Two-layer, and Three-layer FFNNs) performed similarly, with test accuracies around 50.46%–50.55%, but lacked the capacity to model complex relationships. Recurrent models, LSTM and BiLSTM, achieved slightly lower test accuracies (49.58% and 50.55%, respectively), highlighting their limited advantage on this dataset despite their ability to model sequential dependencies. Overall, performance across all models was relatively uniform, suggesting challenges in extracting distinct features from the dataset.

4. Discussion

4.1 Optimizer Impact

Adagrad, with its adaptive learning rate, achieved the highest test accuracy (73.28%) and lowest test loss (0.572), enabling robust generalization, consistent with Basarslan and Kayaalp's findings^[1]. Adam showed competitive validation accuracy (74.03%) but higher test loss (0.593), indicating potential overfitting. In contrast, SGD lagged with slower convergence and the lowest test accuracy (56.01%).

4.2 Epoch Influence

Training for 20 epochs provided the best balance, achieving the highest test accuracy (70.71%) and low test loss (0.593), while 50 epochs caused overfitting with high validation loss (0.902) despite 94.19% training accuracy. Shorter durations (5–10 epochs) resulted in lower accuracies, highlighting ~20 epochs as optimal for this task^[1].

4.3 Model Comparison

CNNs excelled in capturing localized patterns, achieving the highest accuracy (50.86%). This supports findings by Islam et al.^[2], who demonstrated the efficacy of CNNs in leveraging n-gram features for sentiment classification. Recurrent models like LSTM and BiLSTM offered no significant advantage, reflecting limited sequential dependencies in the IMDB dataset^[2]. Feedforward networks performed adequately but lacked the capability to model complex relationships, underscoring CNNs' suitability for sentiment analysis in this context.

5. Conclusion

Adagrad achieved the best performance (73.28% accuracy, 0.572 loss), with 20 epochs balancing generalization and overfitting (70.71% accuracy, 0.593 loss). CNNs outperformed other models (50.86% accuracy), effectively capturing localized patterns, while recurrent and feedforward models showed limited advantages.

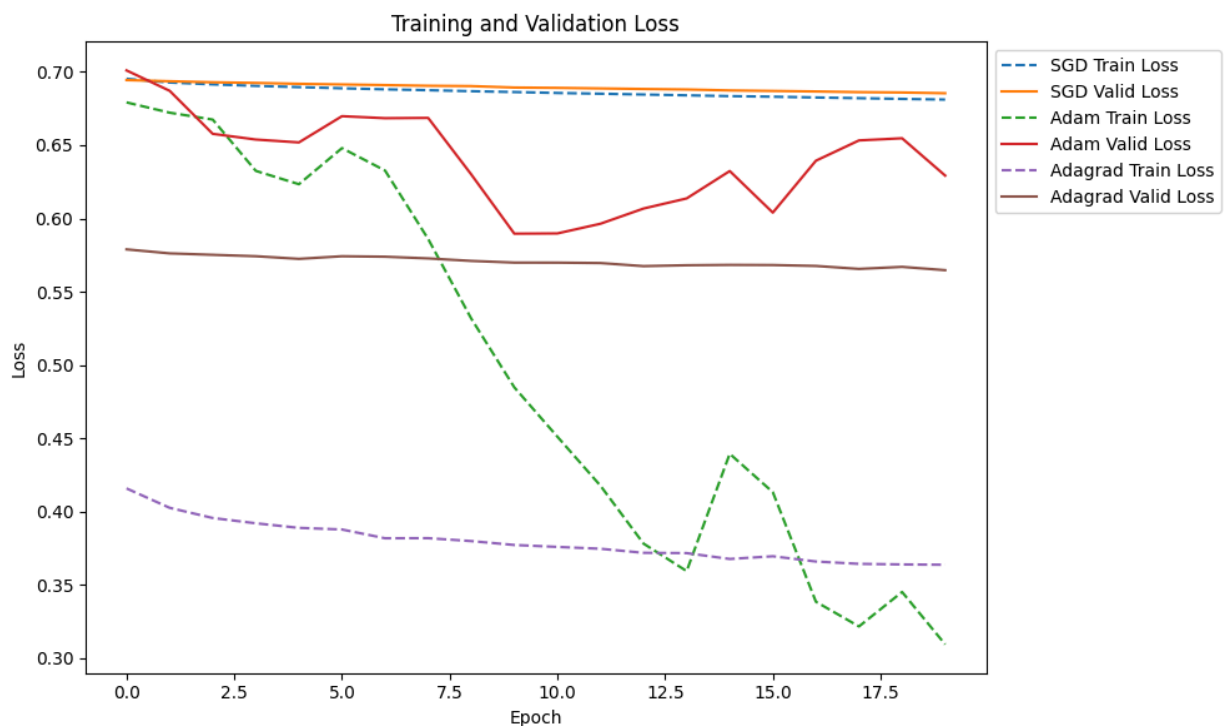
References

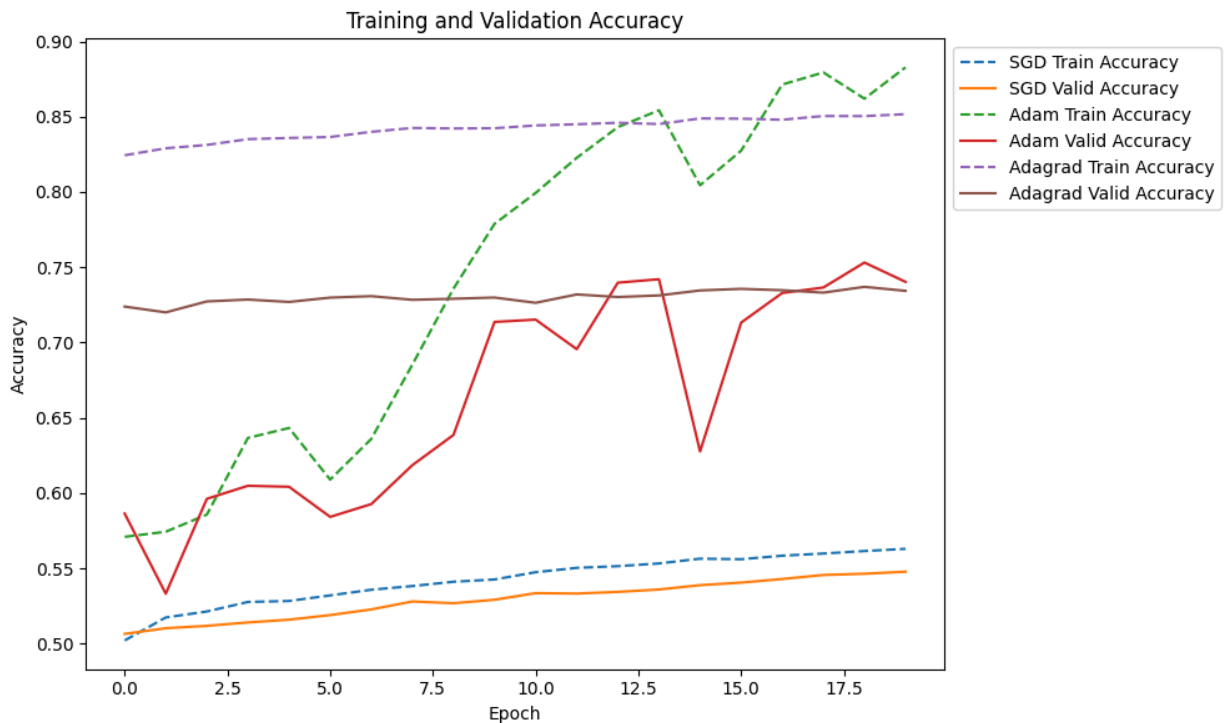
1. M. S. Basarslan and F. Kayaalp, "Sentiment Analysis with Various Deep Learning Models on Movie Reviews," 2022 International Conference on Artificial Intelligence of Things (ICAIoT), Istanbul, Turkey, 29-30 Dec. 2022, pp. 1–6. DOI: 10.1109/ICAIoT57170.2022.10121745.
2. M. T. Islam, F. Parvin, S. A. Sazan, and T. B. Amir, "Comparative Analysis of Sentiment Classification on IMDB 50k Movie Reviews: A Study Using CNN, LSTM, CNN-LSTM, and BERT Models," 2024 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON), Rajshahi, Bangladesh, 12-13 Sept. 2024, pp. 1–8. DOI: 10.1109/PEEIACON63629.2024.10800035.

Appendix

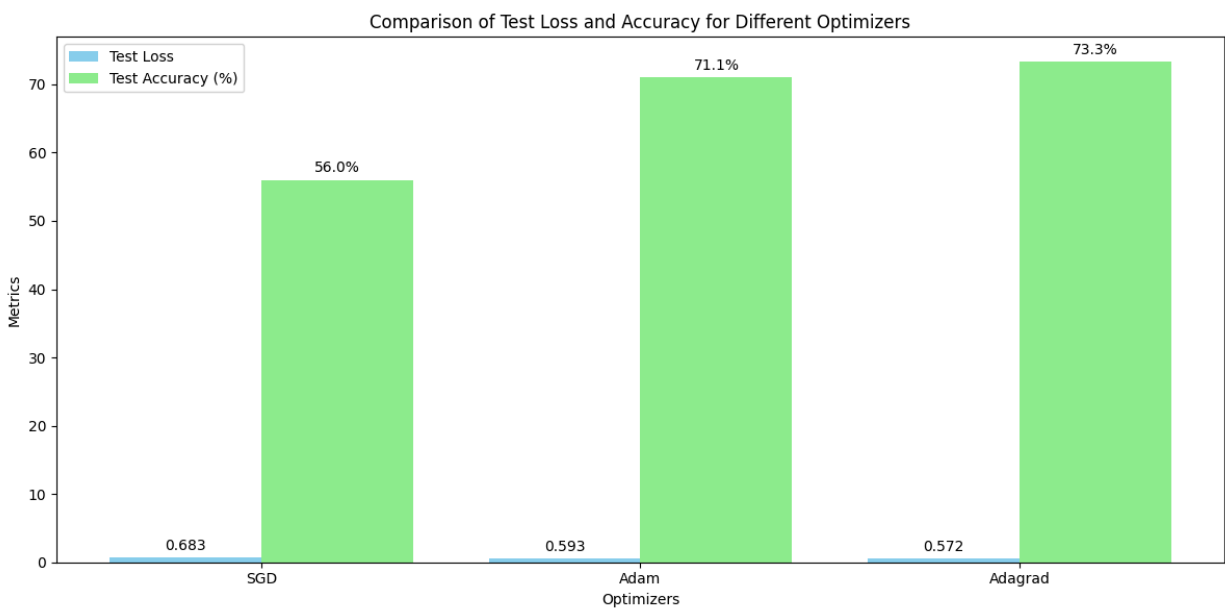
Optimizer Comparison

- Loss and Accuracy:
 - Evaluated the performance of SGD, Adam, and Adagrad optimizers.



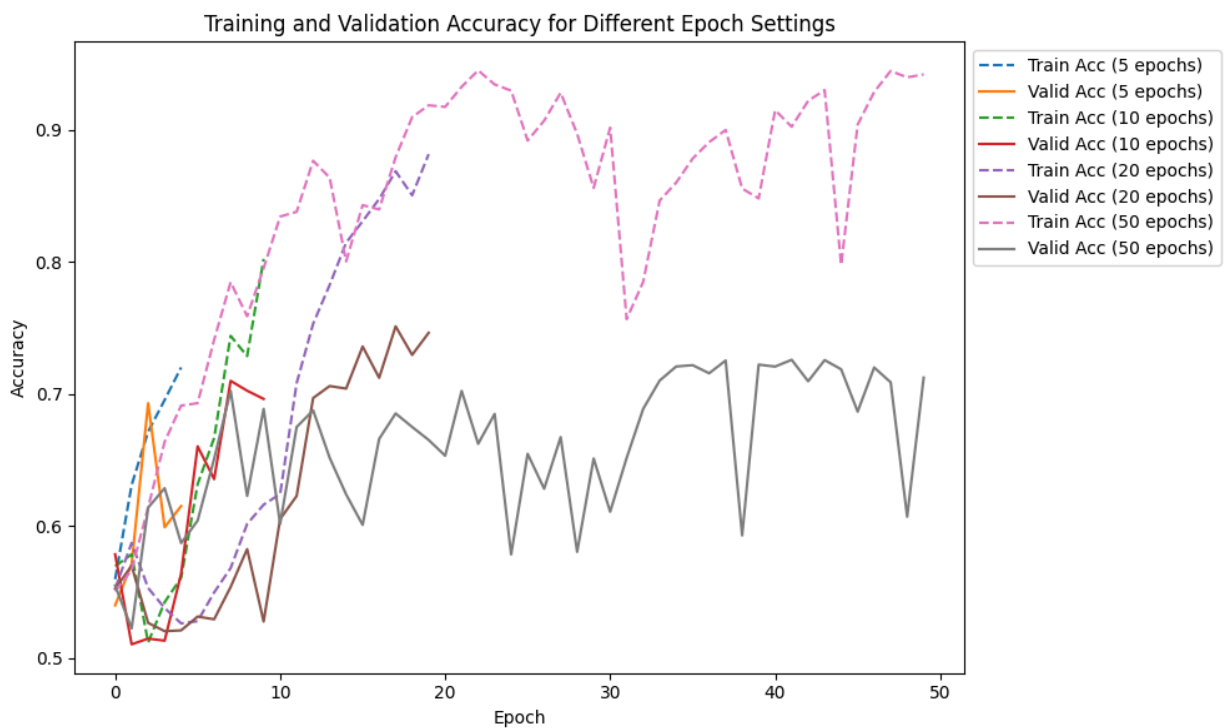
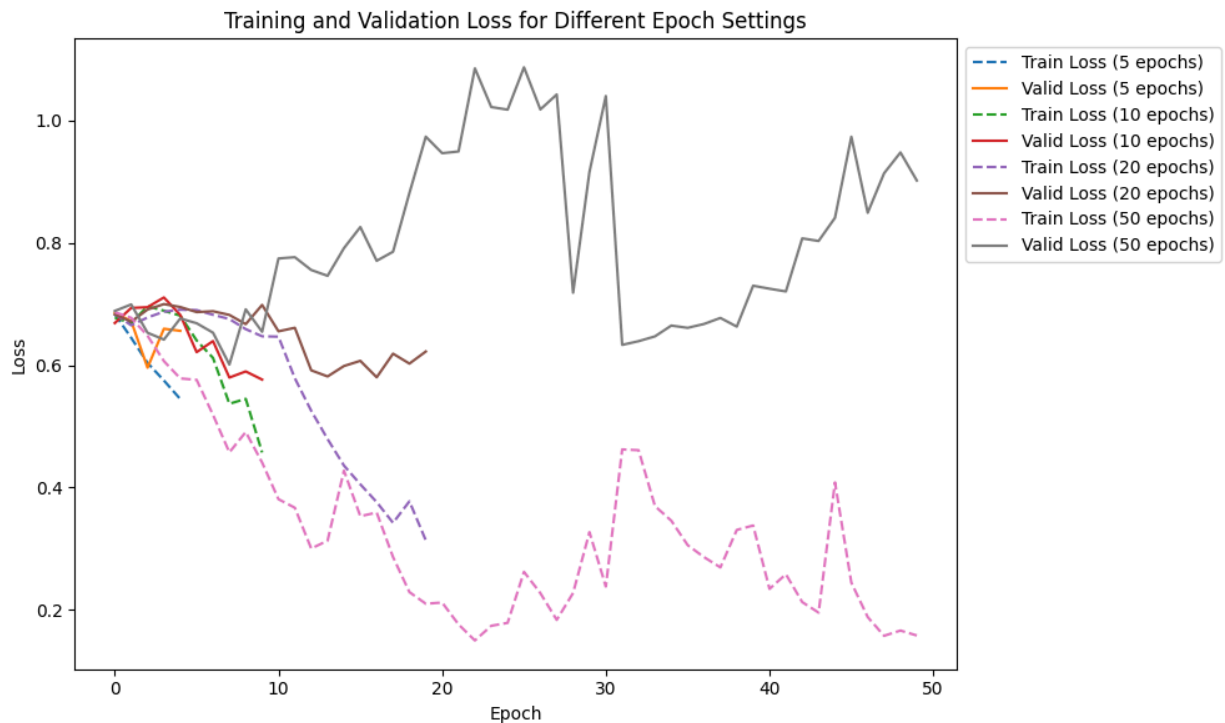


- Comparison:
 - Visualizes test loss and accuracy for the optimizers.

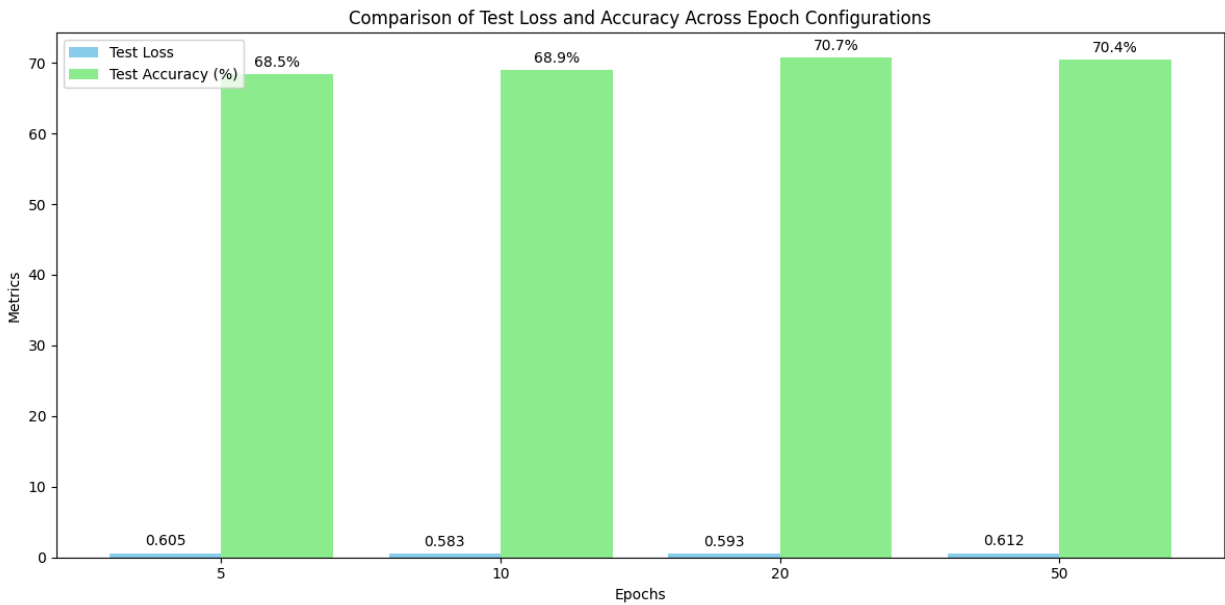


Epoch Variation

- Loss and Accuracy:
 - Explores the effect of different epoch configurations (5, 10, 20, 50).

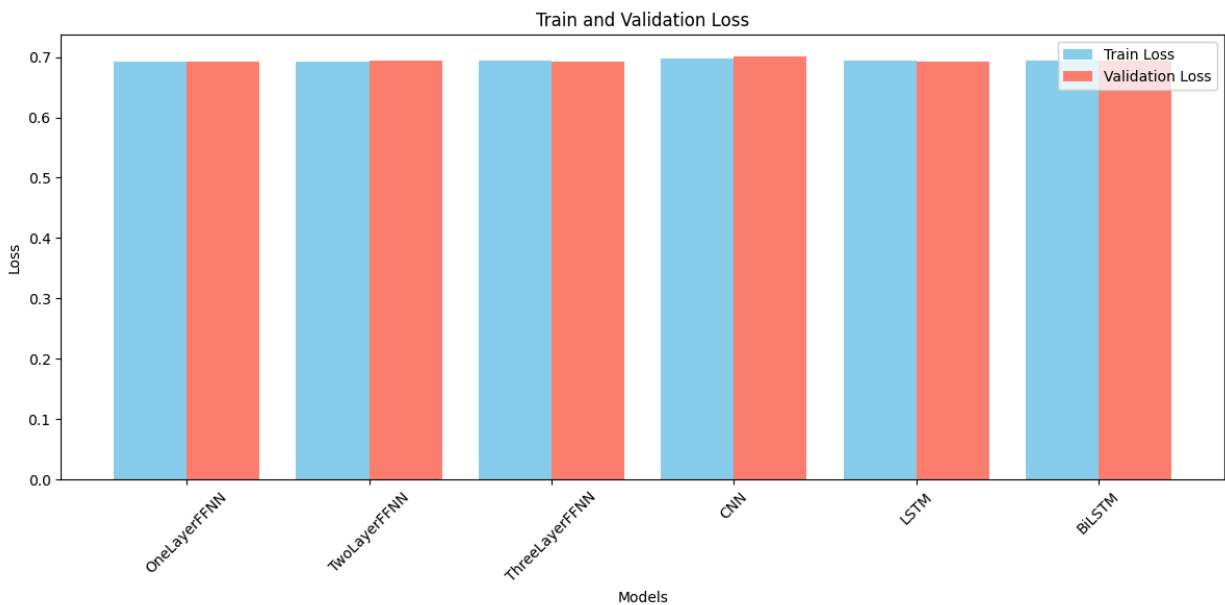


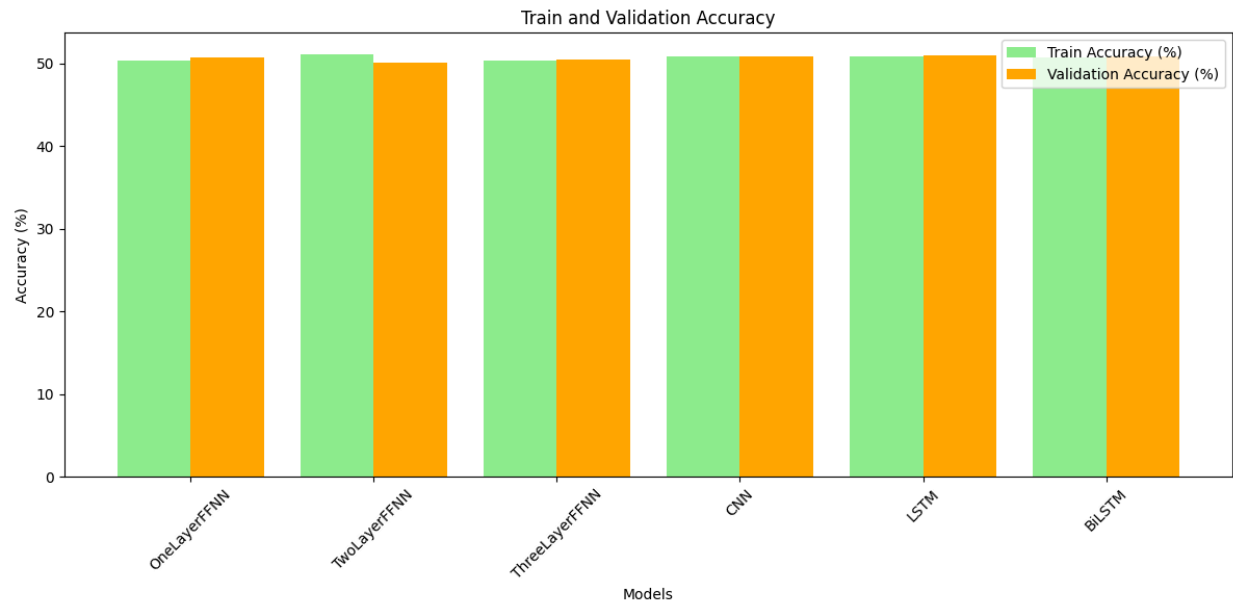
- Comparison:
 - Summarizes the test loss and accuracy across different epochs.



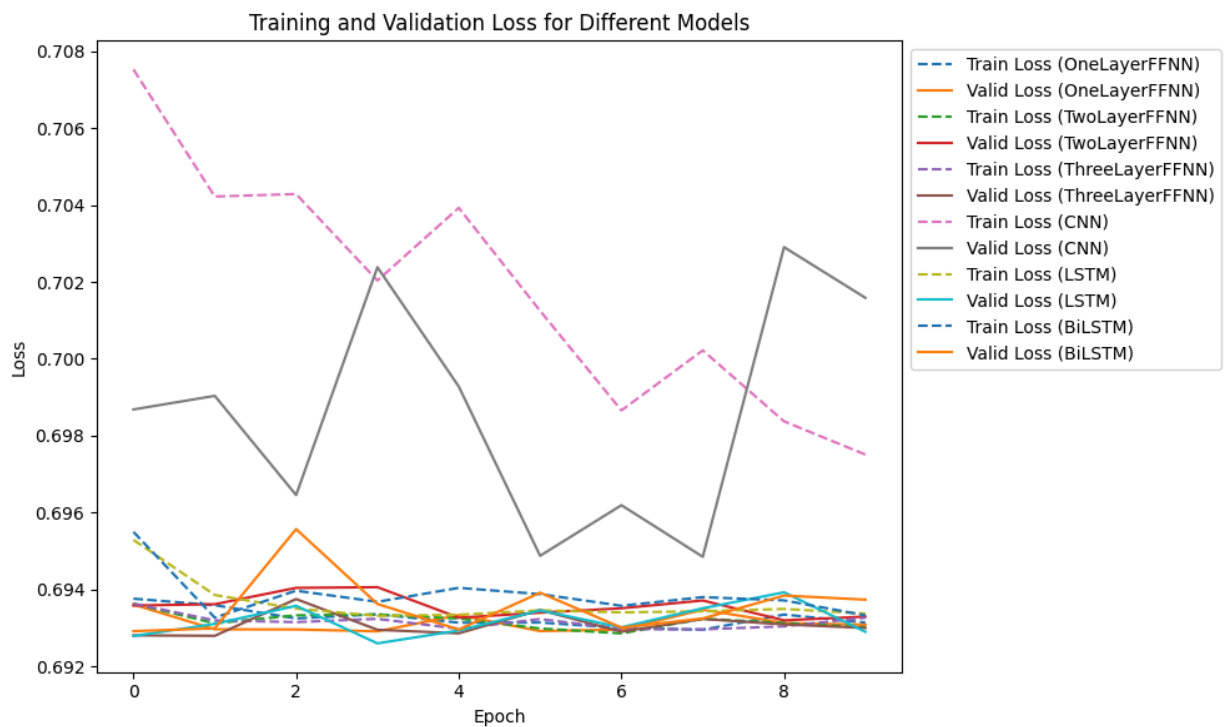
Model Comparison

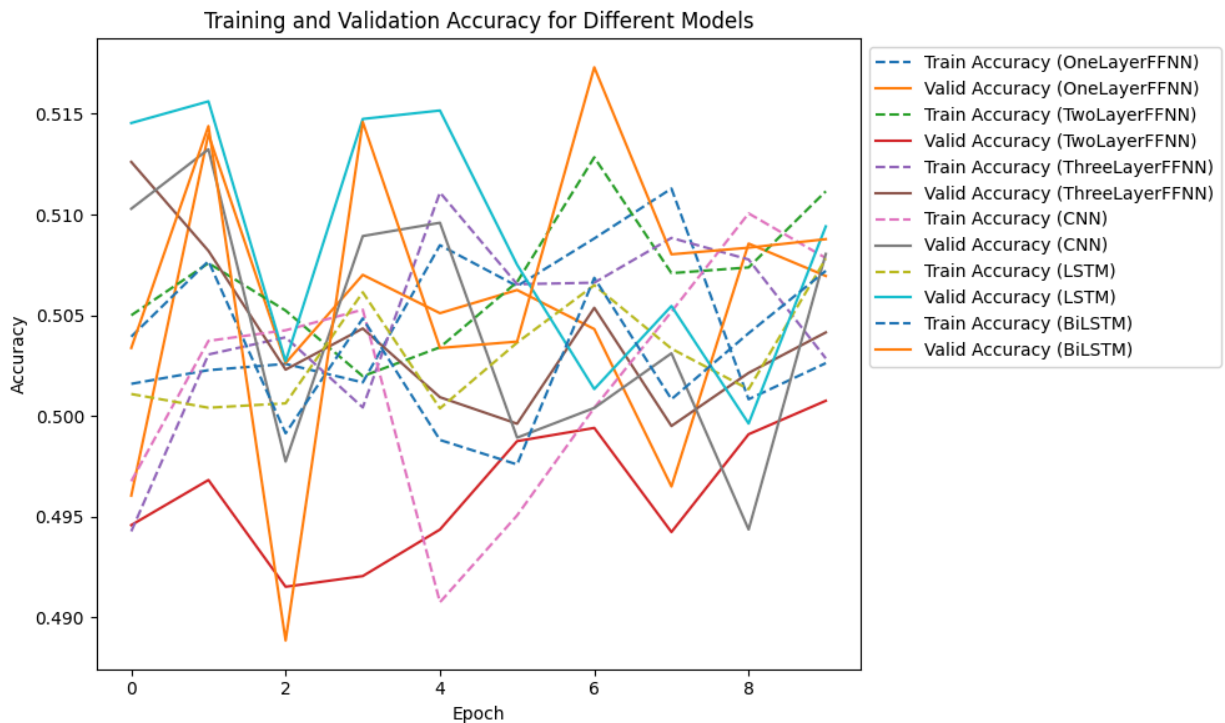
- Loss and Accuracy:
 - Compares performance of six model architectures (One-layer FFNN, Two-layer FFNN, Three-layer FFNN, CNN, LSTM, BiLSTM).





- Training Trends:
 - Highlights training and validation loss/accuracy trends for the models.





- Comparison:
 - Summarizes test loss and accuracy for each model.

